

Matrices

4.3 Name of matrices and their meaning

1. Square Matrices:

Number of Rows = Number of Columns

Example: $P = \begin{bmatrix} E_1 & E_2 & E_3 \\ E_1' & E_2' & E_3' \\ E_1'' & E_2'' & E_3'' \end{bmatrix}_{3 \times 3}$, Here, $m = n$

A^T (Transpose):

Let $A = \begin{bmatrix} E_1 & E_2 & E_3 \\ E_4 & E_5 & E_6 \\ E_7 & E_8 & E_9 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} E_1 & E_4 & E_7 \\ E_2 & E_5 & E_8 \\ E_3 & E_6 & E_9 \end{bmatrix}$

We can write a matrix as: $A = [a_{ij}]_{m \times n}$

For example: $a_{11} = E_1$

$$a_{12} = E_2$$

$$a_{31} = E_7 \text{ etc.}$$

$$(a) (A^T)^T = A$$

$$(b) (AB)^T = B^T A^T$$

$$(c) (A+B)^T = A^T + B^T$$

2. Symmetric Matrix: $(A^T) = A$

$$A = \begin{bmatrix} 20 & 120 & 200 \\ 120 & 10 & 150 \\ 200 & 150 & 30 \end{bmatrix}, \quad A^T = \begin{bmatrix} 20 & 120 & 200 \\ 120 & 10 & 150 \\ 200 & 150 & 30 \end{bmatrix} = A$$

$\therefore A$ is a symmetric matrix.

3. Skew Symmetric Matrix: $(A^T) = -A$

$$B = \begin{bmatrix} 0 & 1 & -3 \\ -1 & 0 & -2 \\ 3 & 2 & 0 \end{bmatrix}, \quad B^T = \begin{bmatrix} 0 & -1 & 3 \\ 1 & 0 & 2 \\ -3 & -2 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & 1 & -3 \\ -1 & 0 & -2 \\ 3 & 2 & 0 \end{bmatrix}$$

$$\therefore B^T = -B$$

$\therefore B$ is a skew-symmetric matrix.

In a skew-symmetric matrix, the diagonal elements will always be equal to zero.

Proof: $A = [a_{ij}]_{n \times n}$

$$A^T = -A$$

$$[a_{ji}]_{n \times n} + [a_{ij}]_{n \times n} = 0$$

$$[a_{ji}] + [a_{ij}] = 0$$

for diagonal elements, $i = j$,

$$\therefore [a_{ii}] + [a_{ii}] = 0 \Rightarrow 2[a_{ii}] = 0$$

$$\Rightarrow [a_{ii}] = 0$$

So, diagonal elements of the skew-symmetric matrix are always zero.

4. Orthogonal Matrix: $A^T A = I \Rightarrow A^T = A^{-1}$

Also the sum of the square of elements of any row = 1.

$$\text{If } \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \text{ is orthogonal } \Rightarrow \begin{aligned} a^2 + b^2 + c^2 &= 1 \\ d^2 + e^2 + f^2 &= 1 \\ g^2 + h^2 + i^2 &= 1 \end{aligned}$$

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

$$B = \frac{1}{2}(A + A^T), C = \frac{1}{2}(A - A^T) \Rightarrow A = B + C$$

$$\text{Now, } B^T = \frac{1}{2}(A^T + A) = B, \therefore B \text{ is symmetric}$$

$$C^T = \frac{1}{2}(A^T - A) = -C, \therefore C \text{ is skew-symmetric}$$

So A matrix can be represented in its symmetric and skew-symmetric parts.

Question: If $P = \begin{bmatrix} a & -2b \\ b & -2b^2 \end{bmatrix}$, then what will be the symmetric part?

$$\text{Solution : } P = \begin{bmatrix} a & -2b \\ b & -2b^2 \end{bmatrix} = \begin{bmatrix} a^2 & -2ab \\ ba & -2b^2 \end{bmatrix}$$

$$\text{For symmetric part } B = \frac{1}{2}(P + P^T) = \frac{1}{2} \left\{ \begin{bmatrix} a^2 & -2ab \\ ba & -2b^2 \end{bmatrix} + \begin{bmatrix} a^2 & ab \\ -2ab & -2b^2 \end{bmatrix} \right\} = \frac{1}{2} \begin{bmatrix} 2a^2 & -ab \\ -ab & -4b^2 \end{bmatrix}$$

$$B = \begin{bmatrix} a^2 & -ab/2 \\ -ab/2 & -2b^2 \end{bmatrix} = \text{Symmetric part of } P.$$

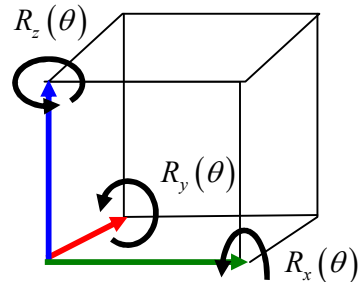
5. Hermitian Matrix: $A^\dagger = A$

$$\text{Example: } A = \begin{bmatrix} 4 & 1-3i \\ 1+3i & 7 \end{bmatrix}$$

$$\text{Take transpose } \Rightarrow A^T = \begin{bmatrix} 4 & 1+3i \\ 1-3i & 7 \end{bmatrix},$$

$$\text{Then take the conjugate } \Rightarrow A^\dagger = \begin{bmatrix} 4 & 1-3i \\ 1+3i & 7 \end{bmatrix}$$

$$\Rightarrow A^\dagger = A \therefore A \rightarrow \text{Hermitian Matrix}$$



6. Skew Hermitian Matrix : $A^\dagger = -A$

Example: $A = \begin{bmatrix} 3i & 2+i \\ -2+i & -i \end{bmatrix}$

$$A^\dagger = \begin{bmatrix} -3i & -2-i \\ 2-i & i \end{bmatrix} = -\begin{bmatrix} 3i & 2+i \\ -2+i & -i \end{bmatrix}, \quad A^\dagger = -A$$

$\therefore A \rightarrow$ Skew Hermitian matrix

7. Unitary Matrix: $A^\dagger A = I$ or $A^\dagger = A^{-1}$

8. Triangular Matrix: $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$ = Principal diagonal

If all elements lie below the principal diagonal are zero $\Rightarrow$ Upper triangular matrix

And if all elements lie above principle diagonal are zero $\Rightarrow$ Lower triangular Matrix

9. Diagonal Matrix:

Example: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

At least one of the Diagonal terms should be non-zero. (We are talking about principle diagonal). Also, all other elements are zero.

10. Unit matrix:

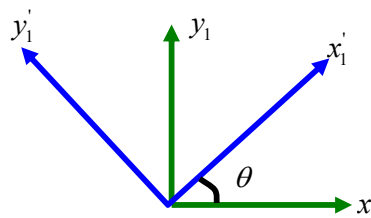
Example: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Principle diagonal has 1's and all other elements are zero. Also known as Identity matrix.

11. Rotation Matrix in 2D:

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x'_1 \\ y'_1 \end{bmatrix}$$



Rotation Matrix in 3D:

$+\theta \rightarrow$ For anticlockwise direction

$-\theta \rightarrow$ For clockwise direction

$$A_x(\pm\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \mp\sin\theta \\ 0 & \pm\sin\theta & \cos\theta \end{bmatrix}$$

$$A_y(\pm\theta) = \begin{bmatrix} \cos\theta & 0 & \pm\sin\theta \\ 0 & 1 & 0 \\ \mp\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$A_z(\pm\theta) = \begin{bmatrix} \cos\theta & \mp\sin\theta & 0 \\ \pm\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

